

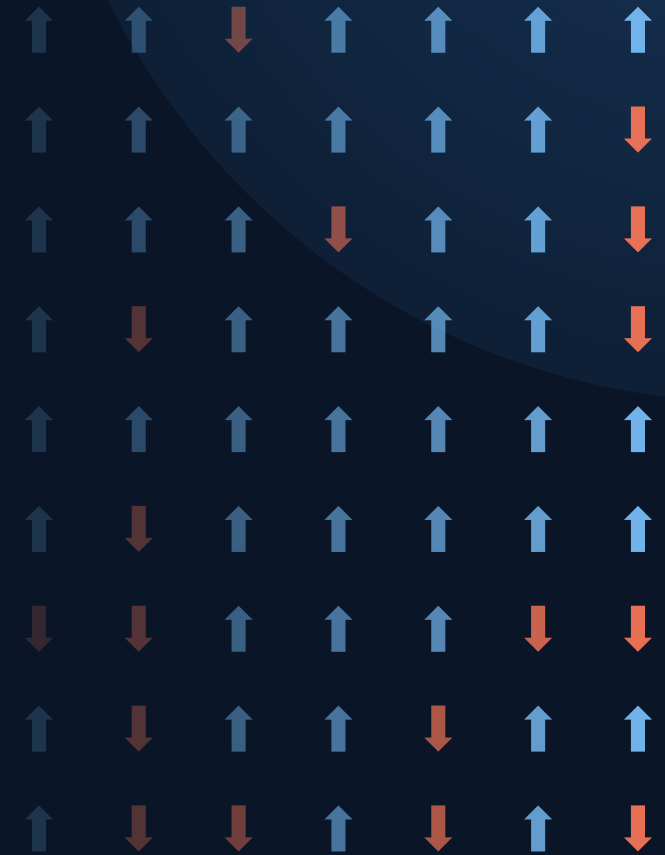
Measuring the External Field

Can we measure how hard a news article pushes on what people think?

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each arrow is one person's opinion



Physics treats media as a push, h , on opinion. Nobody measures h

$$m = \tanh(\beta J m + h)$$

 m

magnetisation
= average opinion, -1 to +1

 J

coupling
= pull toward neighbours

 β

inverse temperature
= how firmly people follow

 h

external field
= media's push on everyone



Question: can h be measured from the content itself?

Next: what testing the tools changed →



Testing the tools changed five assumptions

WE PLANNED

❌ Measure the amygdala

❌ Score as a ratio

❌ Outrage stands out most

❌ Measure α , media's strength

❌ A manipulation detector

WE FOUND

✅ Model predicts the brain surface only, so we used surface areas

✅ Ratio undefined for 69 of 400, so we used a difference

✅ Fear stands out; outrage lifts both areas equally

✅ Data can't pin α , so we derived its minimum

✅ Word counting 0.98 vs our 0.63, so it's a measurement

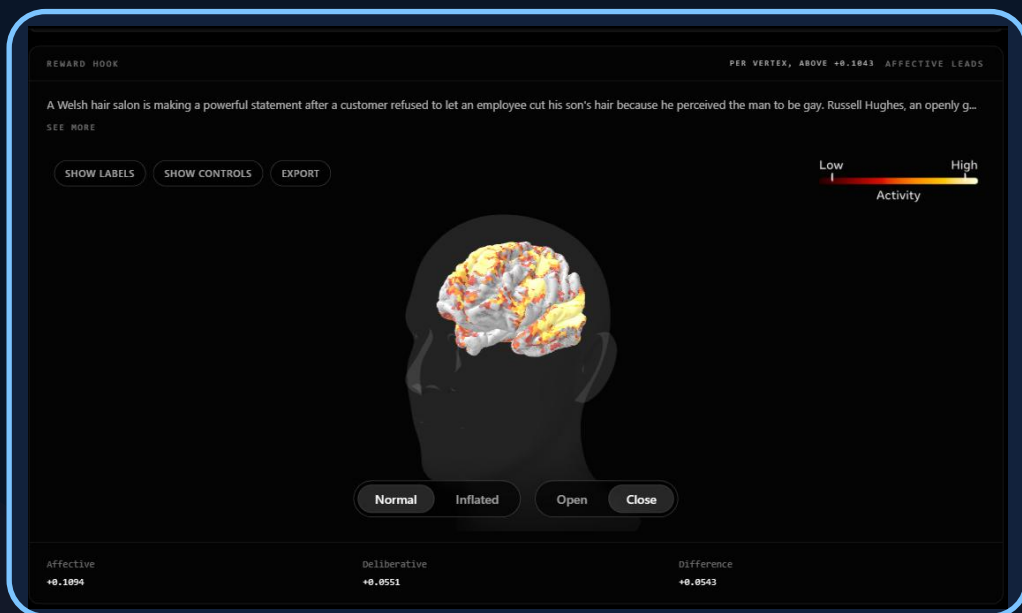
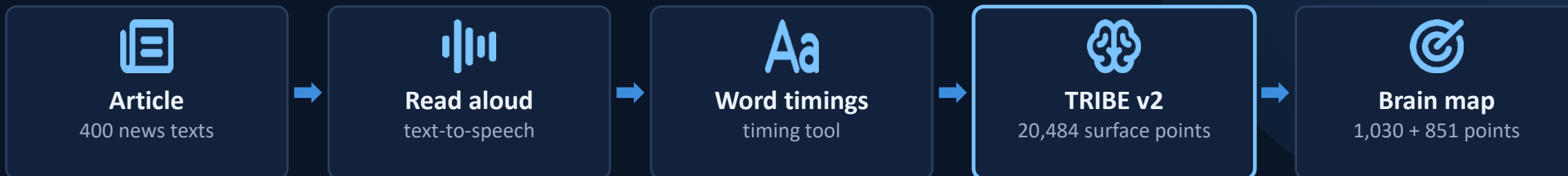
All five are in the approved amendment.

Next: how the measurement works →



THE METHOD

How one article becomes one number



THE SCORE, X

$$X = \bar{A}_{\text{emotional}} - \bar{A}_{\text{deliberate}}$$



$\bar{A}$ = average predicted activity; standardised, no unit · nobody scanned



TRY THE INSTRUMENT LIVE

monarch-4iy.pages.dev

code and data: github.com/brn-mwai/monarch

One article: emotional +0.109 – deliberate +0.055 = score +0.054, highest of 400

Next: the 400 articles →



400 articles, four groups, matched in length

100

Fear-activating
fake news (ISOT)

100

High outrage
partisan news (SemEval)

100

Reward hook
clickbait (Webis)

100

Neutral
medical + real news

162–167

average words per article,
lowest to highest group

0.027

smallest η^2 400 articles can
detect, set before scanning

× 2

every article
scanned twice

All 400, with scores: monarch-4iy.pages.dev/corpus

Next: reading the corpus page →



Reading the corpus page: what each number means

COLUMN → MEANING

Emotional

average predicted activity, 1,030 emotional points

Deliberate

average predicted activity, 851 reasoning points

Score

Emotional – Deliberate = X

Pre-scan label

label from its source, given before any scan

Words

article length

All 400 scores: -0.070 to +0.054, spread $\Delta X = 0.124$

Units: standardised activity, no physical unit

AVERAGE SCORE X PER GROUP

Fear-activating

-0.002

Reward hook

-0.013

High outrage

-0.018

Neutral

-0.019

0

All four below 0: the deliberate areas respond more in every group.

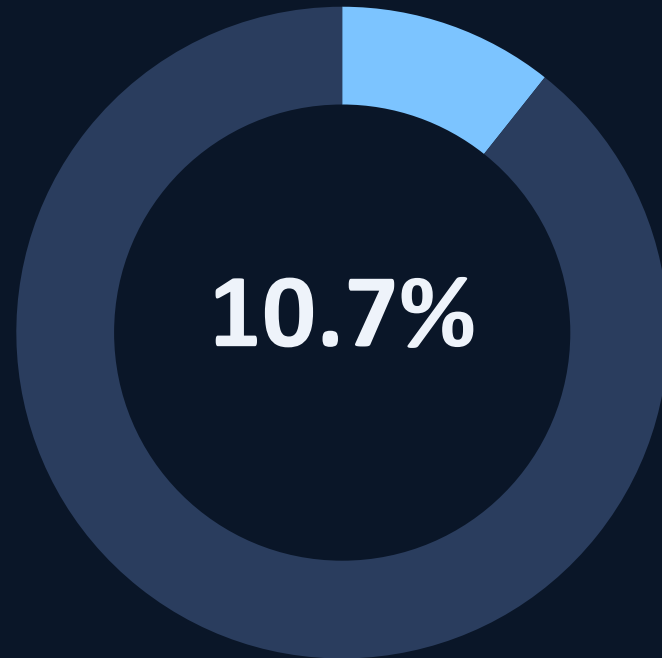
Live: monarch-4iy.pages.dev/corpus

Next: are these differences real? →



THE RESULTS

Result 1: the four groups differ, $\eta^2 = 0.107$



10.7% of the variation in the score is due to the group

$$\eta^2 = \frac{SS_{\text{between groups}}}{SS_{\text{total}}} = 0.107$$

η^2 ON EACH RUN (one-way ANOVA), AGAINST THE SMALLEST WE COULD DETECT



F = 15.779

between-group spread ÷ within-group spread

p = 1.0×10^{-9}

probability of F this large if all 4 groups were equal

SS = sum of squares. The other 89.3% is article to article.

Next: which group drives it? →

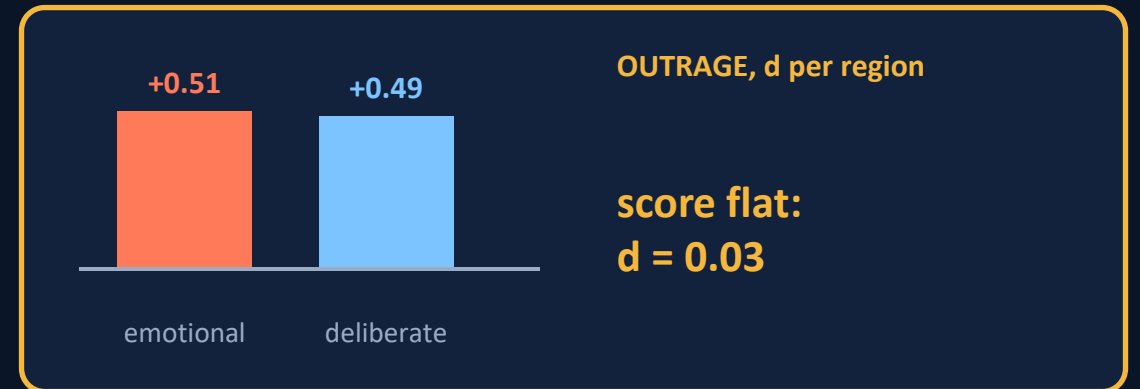
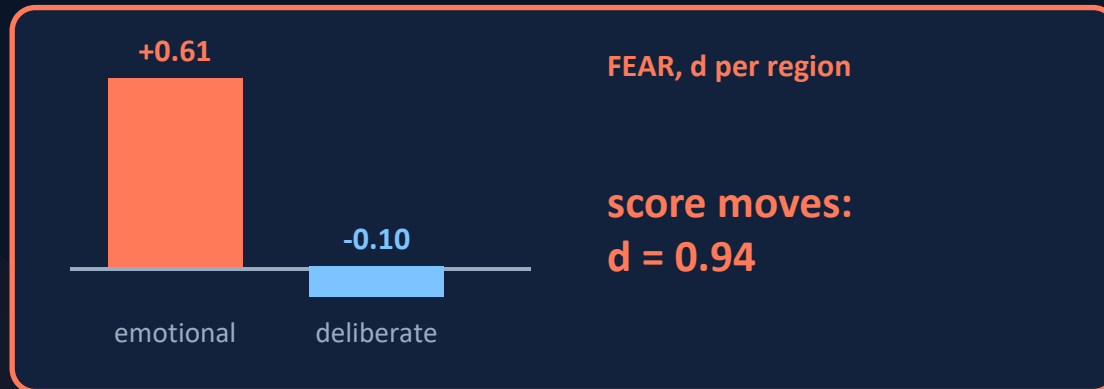
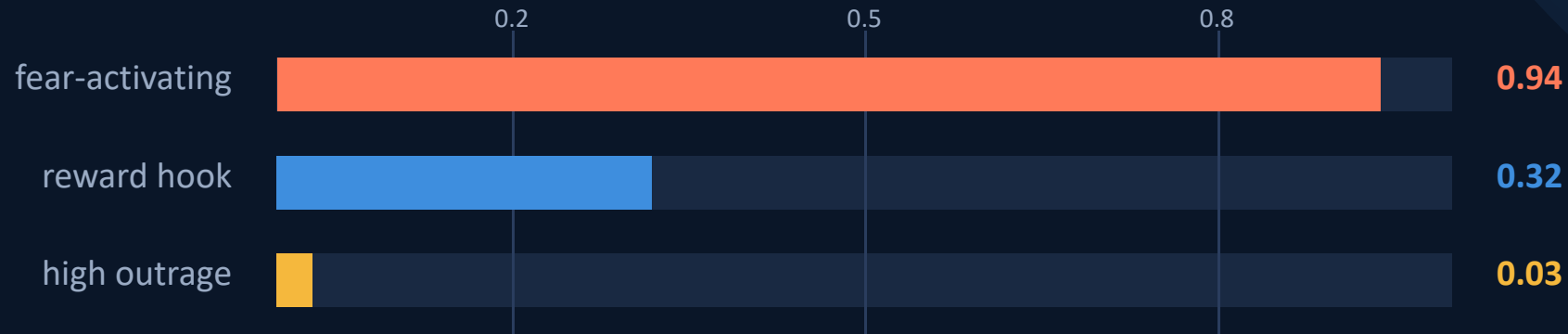


THE RESULTS

Result 2: fear drives it, d = 0.94; outrage lifts both sides equally

$$d = \frac{\bar{X}_{\text{group}} - \bar{X}_{\text{neutral}}}{s}$$

d = distance from neutral in standard deviations (s). Reference marks 0.2, 0.5, 0.8.



Next: can we trust the scores? →



Result 3: scores repeat for groups, not single articles

0.8725

ICC: agreement between two runs



0 = none

1 = identical

51 of 400

changed sign between runs (12.8%)



• observed 51 | noise predicts 55 (95%: 44 to 67)

Meaning: flipped articles average $|X| = 0.003$ against 0.022 for the rest, so we report group averages only.

Next: does the score detect manipulation? →



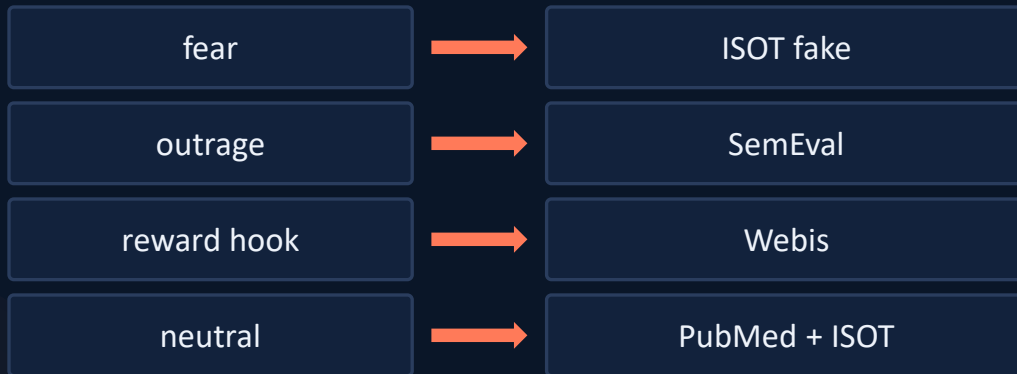
Honest check: word counting beats our score, 0.98 vs 0.63

AUC = chance a random manipulative article scores above a random neutral one



sentiment 0.54

WHY: EACH GROUP CAME FROM ONE SOURCE



66%

of articles: words identify
the source (chance 25%)

So our score is a
measurement for the
physics, not a detector.

Next: does the brain model match real brains? →



Does the brain model match real brains? $r = +0.028$, $p = 0.048$

Public brain scans of 4 people watching Friends (Algonauts 2025). r = correlation: 0 = no match, 1 = perfect.



30%

of the clip ceiling
($0.028 \div 0.096$)

$p = 0.048$

against 61 time-shifted copies;
cut-off 0.05

3 bugs

found and fixed along the way

Meaning: the sign is positive (+0.028). The full-episode run is next, in Paper 3.

Next: the physics →



The minimum push media would need: $\alpha \geq 4.29$

$$h = \alpha X \Rightarrow \alpha \geq \frac{h_c(\beta J)}{\Delta X}$$

h

external field = media's push on everyone

X

observable = our score for an article

α

coupling constant = push per unit of score

h_c

critical field = smallest push that flips the majority

βJ

reduced coupling = how strongly people copy each other

ΔX

spread = highest – lowest score = 0.124



$$\alpha \geq 0.533 \div 0.124 = 4.29$$

at $\beta J = 2$, where the critical field is 0.533

Next: what it could be used for →



Can it analyse any content, even what children watch?



CAN DO

Score video, audio and text

Compare groups of content

Rate content, never the viewer



NOT YET

Model learned from adult brains

Only news text was tested

1 in 8 single scores flip



WOULD TAKE

Child brain data, with ethics approval

A children's video test set

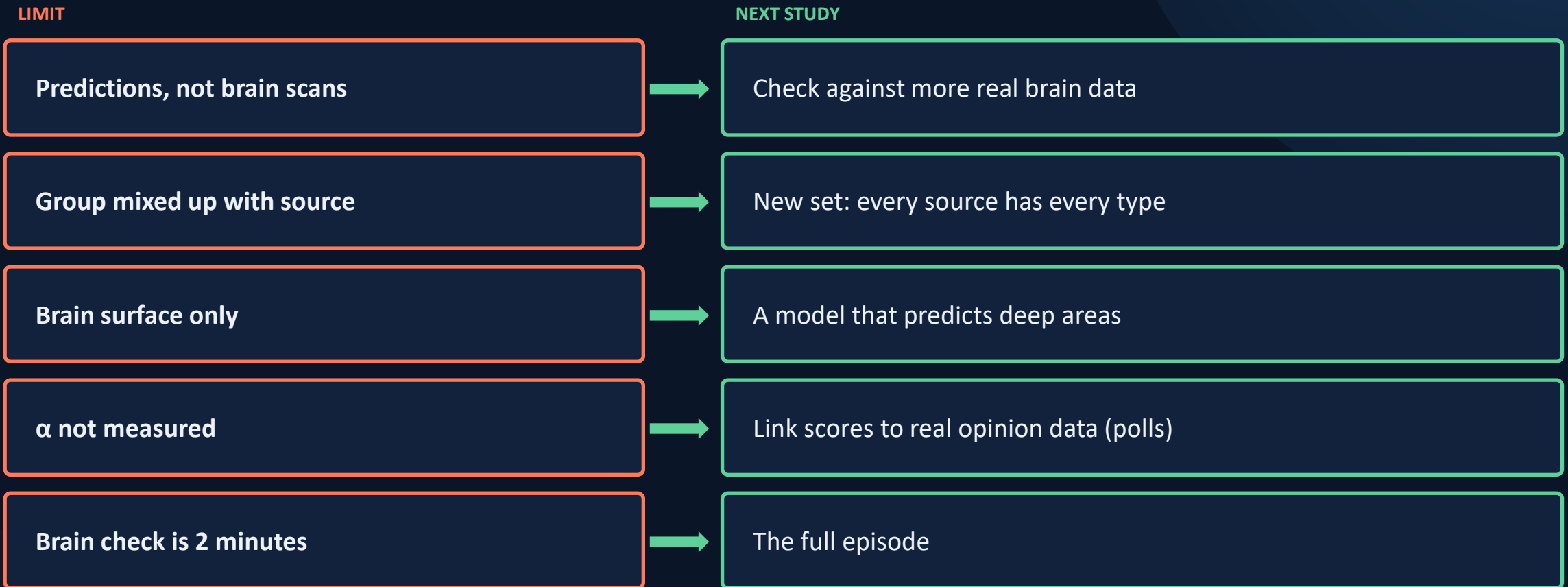
Safeguards against targeting children

Short answer: technically possible, but not yet valid for children.

Next: the limits, and what fixes each →



Each limit points to the next study



Next: where the work goes →



Taking it further: where this research can go



PUBLISH

Paper 1 → Physica A

Paper 2 → Physica A

Paper 3 → Imaging Neuroscience

Preprints first (arXiv)



EXTEND THE SCIENCE

Kenya: 2024 Finance Bill (funding)

Video and audio content

Measure α with opinion data

Children's media, ethically



OPPORTUNITIES

Postgraduate research topic

Neuroscience + media
collaborations

Open tool, already online

Group-level media audits

Next: conclusions →



Conclusions

1 **0.8725**

A field observable can be measured

ICC: agreement between two runs

2 **0.107**

It separates the four groups

η^2 : 10.7% of variation due to group; source mixed in

3 **$\alpha \geq 4.29$**

Media strength has a minimum

at $\beta J = 2$, within the mean-field model

4 **$r = 0.028$**

The brain model matches real brains

positive; 30% of the 0.096 ceiling

Three papers follow. Paper 1 and Paper 2 → Physica A, Paper 3 → Imaging Neuroscience.

Thank you. Questions welcome.